



HR ANALYTICS

HR Analytics Project

Employee Attrition Analysis & Risk Identification

Turning workforce data into clear insights to help identify attrition risk and support better retention decisions.

Team 4

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HR ANALYTICS

Problem Statement

Understanding why employees leave is essential to reducing turnover and protecting performance.

1

What's happening?

Companies face frequent employee attrition.

2

Consequences

High recruitment cost, loss of experience, productivity decline.

3

Core question

Why do employees leave the company?

PROJECT GOAL

Project Objective



Analyze employee behavior

Identify attrition drivers

Detect high-risk employees early

Spot patterns before turnover happens

Support HR decision-making with data

Enable more informed, timely action

Dataset Overview



Demographics

Age, Gender, Education



Job Info

Department, Role, Job Level



Salary

Monthly Income, Salary Hike



Satisfaction

Job, Environment, Work-Life
Balance



Experience

Years at company, Promotion
history



Target

Attrition (Yes / No)

Technical Workflow

A clean, professional four-phase process flowing downward from preparation to reporting.



Data Prep

EDA

Modeling

Dashboards

Python Data Cleaning

Essential preprocessing steps using Python libraries:



Remove unused columns



Drop ID fields



Standardize text values



Remove duplicates



Export cleaned dataset



```
3]: # Drop useless columns  
df = df.drop(columns=["EmployeeCount", "StandardHours", "Over18"])  
  
# Drop id  
df = df.drop(columns=["EmployeeNumber"])  
  
# Clean text  
df["BusinessTravel"] = df["BusinessTravel"].str.replace("_", " ")  
  
# Remove duplicates  
df = df.drop_duplicates()
```



```
4]: # save the file  
df.to_excel("HR_Cleaned_1.xlsx", index=False)
```



Excel Data Transformation

Main Tasks:



Created Age Groups



Created Income Categories



Converted ratings into labels



Prepared data for dashboard visualization

File Home Insert Draw Page Layout Formulas Data Review View Help Power Pivot Table Design

 Cut
 Copy
 Format Painter
Clipboard

Aptos Narrow 11 A[^] A^v
B I U   A

      
Alignment

General
\$

G2 :   

=IF(A2<25,"Under 25",IF(A2<35,"25-34",IF(A2<45,"35-44",IF(A2<55,"45-54","55+"))))

File Home Insert Draw Page Layout Formulas Data Review View Help Power Pivot Table D

 Paste  Cut  Copy  Format Painter	Aptos Narrow 11 A[^] A^v B <i>I</i> <u>U</u>   	           Merge & Center
Clipboard	Font	Alignment

AH2 :   

=IF(S2<5000,"Low",IF(S2<10000,"Mid",IF(S2<15000,"High","Very High")))

Why Data Cleaning Matters?

Impact of Data Preparation:

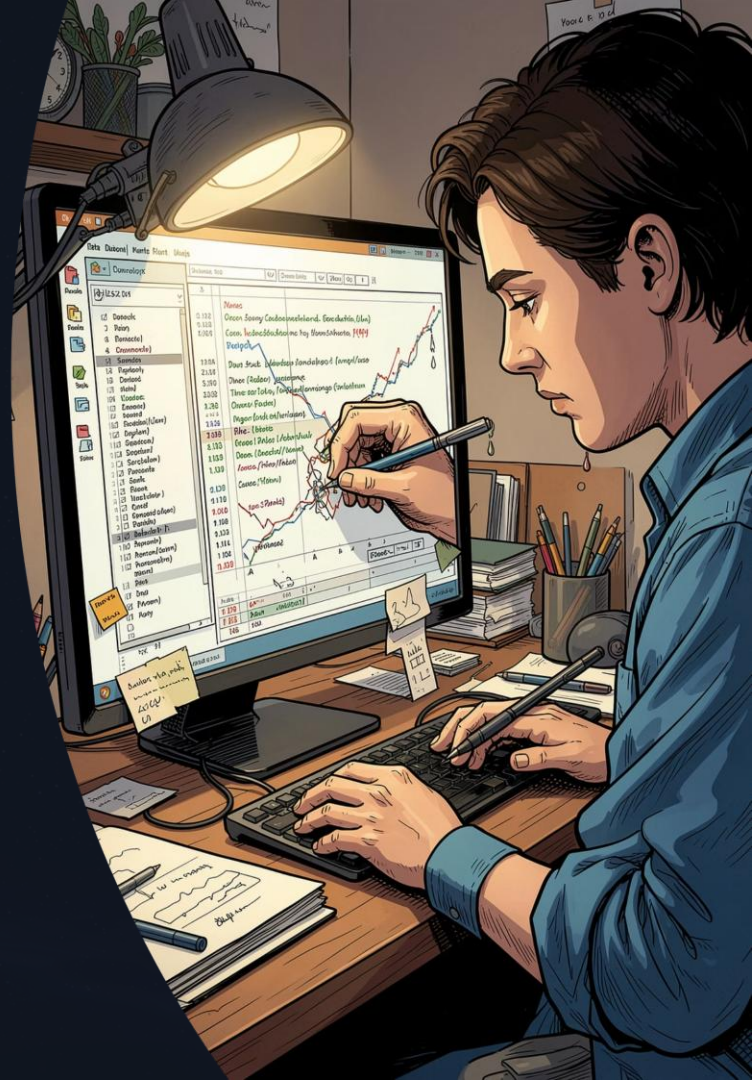
Improved data quality

More accurate analysis

Better visualization

Easier segmentation & insights

Output: Cleaned dataset ready for analysis



SQL Data Analysis

Database queries to extract and analyze HR patterns



Database queries to extract and analyze HR patterns:

- Extract HR insights
- Aggregate workforce data
- Analyze attrition patterns
- Support KPI calculations

Key SQL Operations

WHERE

GROUP BY

COUNT() / AVG()

CASE WHEN

```
SELECT
  department,
  COUNT(*) AS employee_count,
  AVG(salary) AS avg_salary
FROM employees
WHERE status = 'Active'
GROUP BY department;
```

Key Findings from SQL

Key Insights:

- Sales department showed the highest attrition
- Employees under 30 had higher turnover
- Overtime employees were more likely to leave
- Lower job satisfaction increased attrition risk

SQL Supported:

Attrition Analysis

Salary Analysis

Workforce Segmentation

Employee Risk Detection

File Edit View Git Project Tools Extensions Window Help Search

master Execute

Object Explorer

Connect

DESKTOP-QLGH9L7 (SQL Server 17.0.1110.1 - DESKTOP-QLGH9L7)

- Databases
- Security
- Server Objects
- Replication
- Always On High Availability
- Management
- SQL Server Agent
- XEvent Profiler

HR_data.sql - ...l store (55)

```
1 SELECT
2     COUNT(*) AS Total_Employees
3 FROM hr_data;
4 SELECT
5     ROUND(
6         (SUM(CASE WHEN Attrition = 1 THEN 1 ELSE 0 END) * 100.0)
7         / COUNT(*),
8     2) AS Attrition_Rate
9 FROM hr_data;
10 SELECT
11     Department,
12     COUNT(*) AS Attrition_Count
13 FROM hr_data
14 WHERE Attrition = 1
15 GROUP BY Department
16 ORDER BY Attrition_Count DESC;
17 SELECT
18     CASE
19         WHEN Age < 30 THEN 'Under 30'
20         WHEN Age BETWEEN 30 AND 40 THEN '30-40'
21         WHEN Age BETWEEN 41 AND 50 THEN '41-50'
22         ELSE 'Above 50'
23     END AS Age_Group,
24     COUNT(*) AS Attrition_Count
25 FROM hr_data
26
27
28
```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

Connected. (1/1) DESKTOP-QLGH9L7 (17.0 RTM) DESKTOP-QLGH9L7\origin... master 00:00:00 Row: 0, Col: 0 0 rows

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

SQL

SQL

File

Database

Query

Tools

Scripting

Help

Navigator

SCHEMAS

Filter objects

new_schema

sakila

Tables

Views

Stored Procedures

Functions

sys

world

Query 1

Mohamed Nasser Corr and Reg...

Limit to 1000 rows

✱

🔍

📄

🔄

190 (

191 STDDEV(YearWithCurrManager) * STDDEV(Attrition)

192)

193 AS corr_YearWithCurrManager

194 FROM new_schema.`hr\_cleaned\_1(sheet1)`;

195 -- Regression part

196 SELECT

197 -- Age

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Result Grid

Standar_Age	StandardBusinessTravel	StandardDailyRate	StandardDistanceFromHome	StandardEduca
-0.05854571285727213	0.03137013226939844	-0.020833083795354302	0.028655453759570564	-0.01153697441

Administration

Schemas

Information

No object selected

Object Info

Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result 1

Result 2

Result 3

Result 4 ✕

Read Only

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	18:09:40	SELECT (AVG(Age * Attrition) - AVG(Age) * AVG(Attrition)) / (STDDEV(Age) * STDDEV(Attritio...	1 row(s) returned	0.047 sec / 0.000 sec
✓ 2	18:09:40	SELECT -- Age ((AVG(Age * Attrition) - AVG(Age) * AVG(Attrition)) / (AVG(Age * Age) - POW...	1 row(s) returned	0.015 sec / 0.000 sec
✓ 3	18:09:40	SELECT (AVG((JobSatisfaction - stats.avg_js) / stats.std_js) * Attrition) - AVG((JobSatisfac...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 4	18:09:40	SELECT /* Age */ (AVG(((Age - stats.avg_Age) / stats.std_Age) * Attrition) - AVG((Age - stat...	1 row(s) returned	0.063 sec / 0.000 sec

Activate Windows
Go to Settings to activate Windows.

Query Completed

Hypothesis (1)

1. Job Difficulty and Attrition (Positive Relationship) (+)

We assume a positive relationship between job-related difficulties and employee attrition. Variables such as **travel frequency** and **distance between home and work** increase the likelihood of attrition.

2. Adaptation and Stability Factors (Mixed Relationships)

A. Employment History (Positive Effect on Attrition) (+)

Employees who have worked in a higher number of companies (greater job mobility/turnovers) are more likely to experience attrition

B. Stability and Organizational Attachment (Negative Effect on Attrition) (-)

The following factors are expected to reduce the likelihood of attrition:

- Age (older employees tend to be more stable in the labor market)
- Years at the company (higher organizational attachment)
- Performance rating (strong performance increases retention incentives)
- Job involvement

Hypothesis (2)

- Job level (higher positions reduce willingness to take risk and change jobs)
- Job satisfaction
- Relationship satisfaction
- Work–life balance
- Training received in the last year (reflecting both skill development and organizational investment)

3. Financial Incentives (Negative Relationship) (-)

Higher financial incentives are expected to reduce the likelihood of employee attrition (The variables are Monthly Income, Stock Option Level, Percent Salary Hike).

Results (1)

- We conducted **correlation, regression, and standardization analyses** on the dataset.
- As hypothesized at the beginning, all hypotheses were verified and found to be supported by the results.
- We used regression analysis in addition to correlation analysis because we aim to examine **causal relationships and the magnitude of their effects** (i.e., coefficients in the regression table).
- Since **monthly income is measured in thousands**, unlike the other variables, its inclusion in regression without adjustment could distort the results. Therefore, we applied **standardization to all variables** to ensure a balanced and comparable dataset.
- It is noted that the coefficients are relatively small. This may be explained by several factors:
 1. Employee attrition is influenced by many variables. We use 15 variables and there are also beyond those included in this model.
 2. The analysis captures general relationships, which may become stronger when further segmented by variables such as gender, educational background, age groups, and departments.

Results (2)

- According to the results, **travel frequency** appears to be the strongest factor increasing the likelihood of attrition.
- On the other hand, **monthly income** is the most influential financial factor, while **age** and **job level** are the most important factors, related to stability and adaptation, associated with reducing attrition.

	Factors related to attrition	Regression	Correlation	Standardization
1	Distance From Home	0.004	0.078	0.029
2	Travel Frequency	0.040	0.085	0.031
3	No of Companies worked before	0.006	0.043	0.016
4	Age	-0.006	-0.159	-0.059
5	Years at Company	-0.008	-0.134	-0.049
6	Performance Rating	0.003	0.003	0.001
7	Jobs Involvement	-0.067	-0.130	-0.048
8	Job Level	-0.056	-0.169	-0.062
9	Job Satisfaction	-0.035	-0.103	-0.038
10	Relationship Satisfaction	-0.016	-0.046	-0.017
11	Work Life Balance	-0.033	-0.064	-0.024
12	Training Times Last Year	-0.017	-0.059	-0.022
13	Monthly Income	-1.24893E-05	-0.160	-0.059
14	Stock Option Level	-0.059	-0.137	-0.050
15	Percent Salary Hike	-0.001	-0.013	-0.005

Data Model & DAX Measures

Understanding the foundational data model and key analytical measures is crucial for accurate attrition analysis.

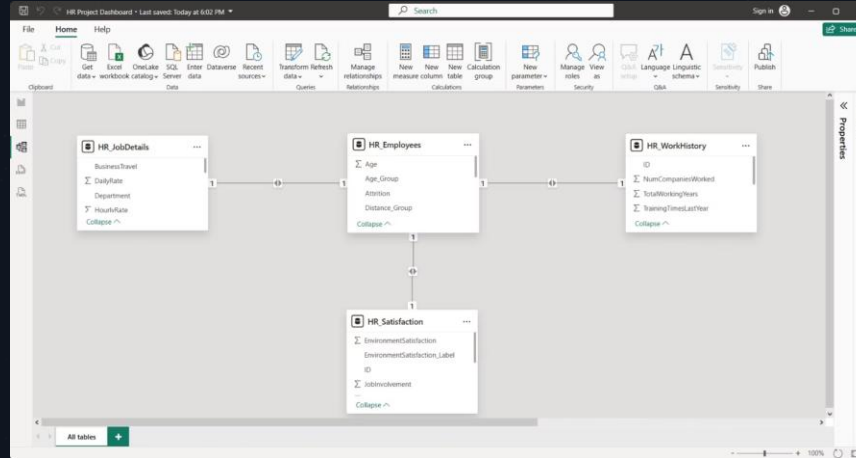


Table Relationships:



Key DAX Measures:

These DAX measures were developed to calculate attrition rate, assess individual employee risk, categorize risk levels, and project the impact of salary increases on attrition reduction.

1. Attrition Rate

```
DIVIDE([Attrition  
Count], [Total  
Employees], 0)
```

2. Attrition Risk Score

```
VAR OverTimeScore =  
IF(MAX(HR_JobDetails[Ov  
erTime]) = "Yes", 30,  
0)  
VAR SatisfactionScore =  
IF(MAX(HR_Satisfaction[  
JobSatisfaction]) <= 2,  
25, 0)  
VAR BalanceScore =  
IF(MAX(HR_Satisfaction[  
WorkLifeBalance]) <= 2,  
20, 0)  
VAR DistanceScore =  
IF(MAX(HR_Employees[Dis  
tanceFromHome]) >= 20,  
15, 0)  
VAR StockScore =  
IF(MAX(HR_JobDetails[St  
ockOptionLevel]) = 0,  
10, 0)  
RETURN OverTimeScore +  
SatisfactionScore +  
BalanceScore +  
DistanceScore + StockScore
```

3. Risk Category

```
IF([Attrition Risk  
Score] >= 51, "High  
Risk",  
IF([Attrition Risk  
Score] >= 21,  
"Medium Risk", "Low  
Risk"))
```

4. Projected Attrition Reduction

```
[Attrition Rate] *  
(1 - 'Salary  
Increase %'[Salary  
Increase % Value] /  
200)
```

Power BI Dashboard Overview

Key Performance Indicators:

1,470

Total Employees

1,233 active

\$6,503

Average Monthly Income

37

Average Age

productive workforce stage

2.69

Avg Previous Companies

Dashboard Features:



Real-time KPI monitoring and trend tracking



Employee segmentation by department, role, and demographics



Interactive filtering for drill-down analysis



Attrition tracking and risk identification

Dashboard Insights — Overview Page

Critical Findings:

16.12%

Attrition Rate

1 out of every 6 employees resigns — a high turnover rate requiring immediate attention.

2.73

Low Employee Satisfaction

Company-wide issue across all departments, not isolated to a single area.

R&D department: 961 employees (largest) HR department: 63 employees (smallest). This makes R&D the most critical department in any business decision.

215

Promotion Gap

No promotion for 5+ years, with highest concentration among Sales Executives (63 employees). Potential source of future attrition.

1.47K

Total Employees

37

Avg Age

6.50K

Avg Monthly Income

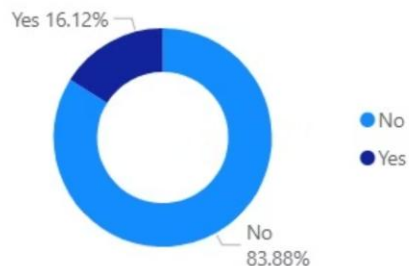
1.23K

Active Employees

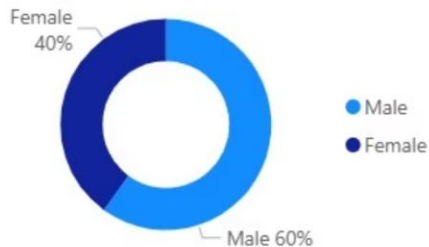
3

Avg Num Companies Worked

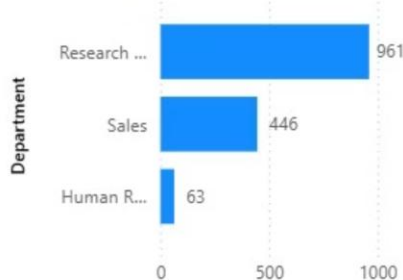
Total Employees by Attrition



Total Employees by Gender



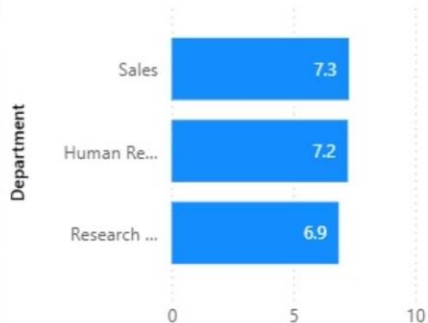
Total Employees by Department



Age_Group

- ☐ Select all
- ☐ +46
- ☐ 18-25
- ☐ 26-35
- ☐ 36-45

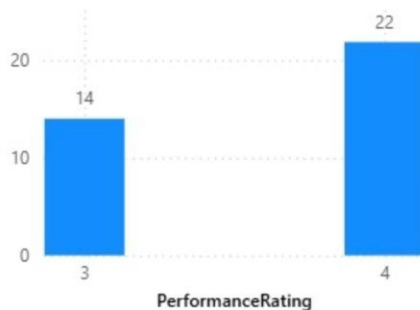
Avg Years At Company by Department



Avg Job Satisfaction by Department



Avg Percent Salary Hike by PerformanceRating



ID	Department	JobRole
46	Research & Development	Research Dir
124	Research & Development	Research Dir
127	Research & Development	Healthcare R
179	Sales	Sales Executi
238	Sales	Manager
426	Research & Development	Manager
702	Sales	Manager
753	Research & Development	Laboratory T
862	Sales	Manager
915	Research & Development	Healthcare R
927	Sales	Sales Executi
938	Research & Development	Manager
977	Research & Development	Manufacturi
585	Research & Development	Manager
654	Research & Development	Manager
923	Research & Development	Manager
1079	Research & Development	Research Dir
1157	Research & Development	Manufacturi
1222	Research & Development	Healthcare R

Dashboard Insights — Attrition & Salary Analysis

Key Findings:

Significant Salary Gap by Role

Managers: \$17,182 vs **Sales**

Representatives: \$2,626

555% salary difference explains
39.8% attrition rate among Sales
Representatives.

Age-Salary Relationship

Salary increases with age until age
40, then declines regardless of
company tenure.

Critical retention issue for
experienced employees over 40.

Stock Options Impact on Attrition

No Stock Options: 24.4% attrition
risk (highest)

Medium Stock Options: 7.6%
attrition risk (lowest)

Financial benefits are key retention
drivers.

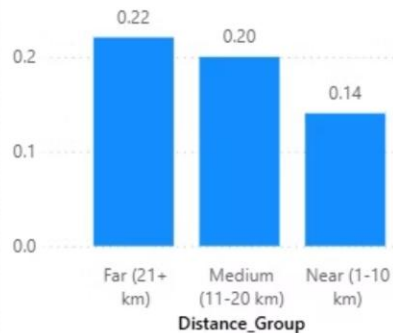
16.12%

Attrition Rate

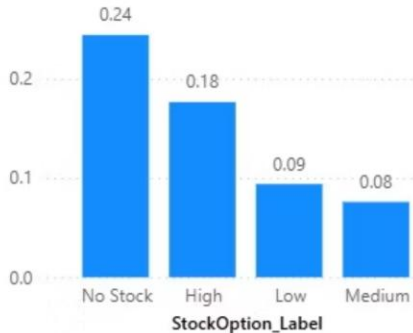
2.73

Avg Job Satisfaction

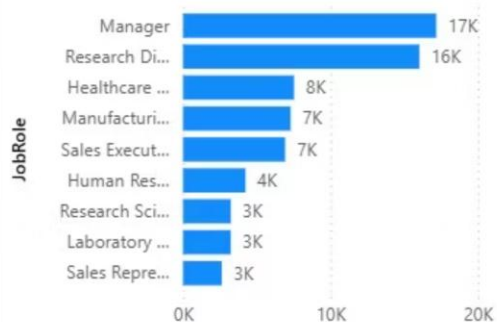
Attrition Rate by Distance_Group



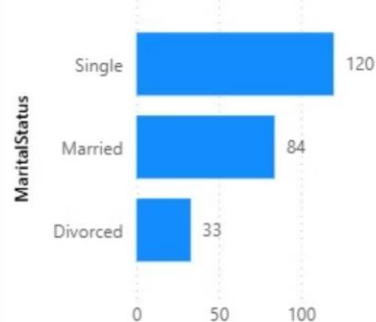
Attrition Rate by StockOption_Label



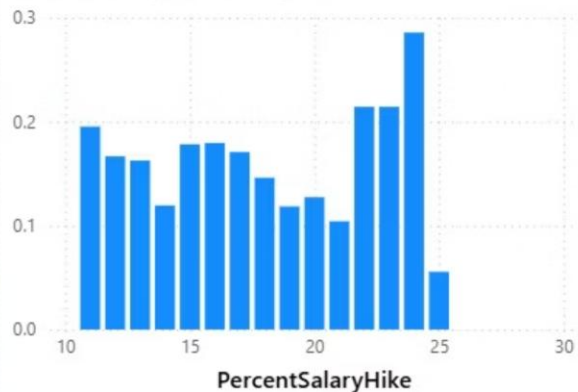
Avg Monthly Income by JobRole



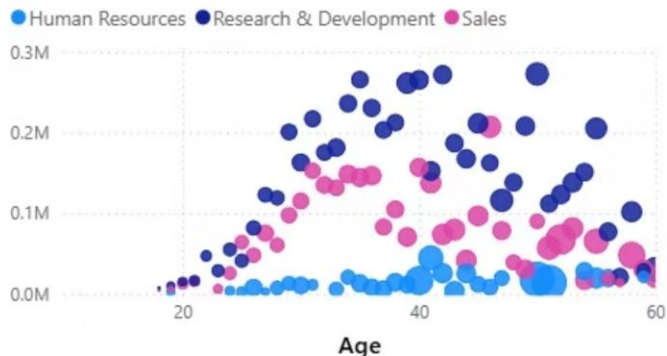
Attrition Count by MaritalStatus



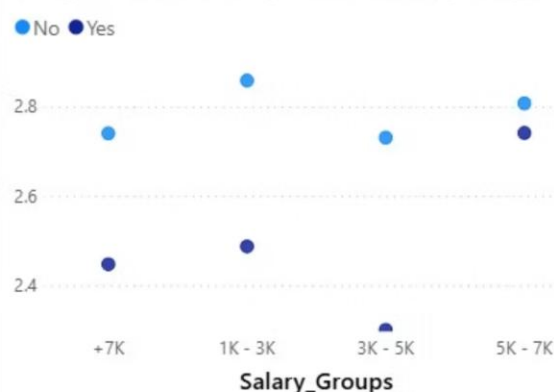
Attrition Rate by PercentSalaryHike



Sum of MonthlyIncome and Average of YearsAtCompany by Department and Age



Average of JobSatisfaction by Attrition and Salary_Groups



Predictive Analysis & Forecasting

Salary Growth Trajectory

Salary increases with overall work experience, rising from \$1,524 to \$6,623 during the first 9 years. Forecast predicts continued growth, reaching up to \$14,730 with extended tenure.

Employee Satisfaction Plateau

Employee satisfaction remains almost constant between 2.59 and 2.92 regardless of years of experience. Indicates company is not improving work environment over time — a structural issue requiring intervention.

Salary Increase Impact Simulation

If salaries increase by 20%:

- Projected income: \$6,503 → \$7,804
- Attrition rate: 16.12% → 15%
- At-risk employees: 237 → 213

Enables management to make data-driven decisions for employee retention.

High-Risk Employee Identification

High-risk employees have simultaneously:

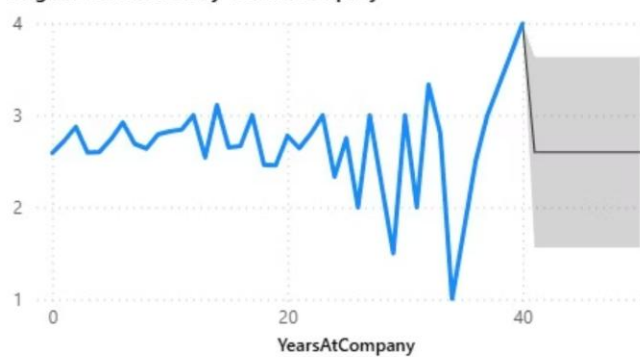
- Overtime work
- Low job satisfaction
- No stock options

The dashboard identifies these employees by name, enabling HR to intervene before resignation.

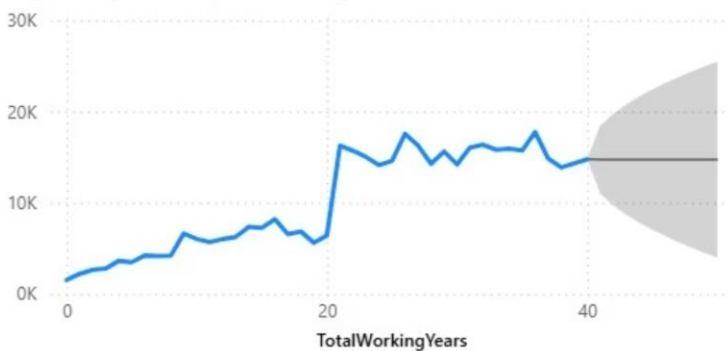
Salary Increase %



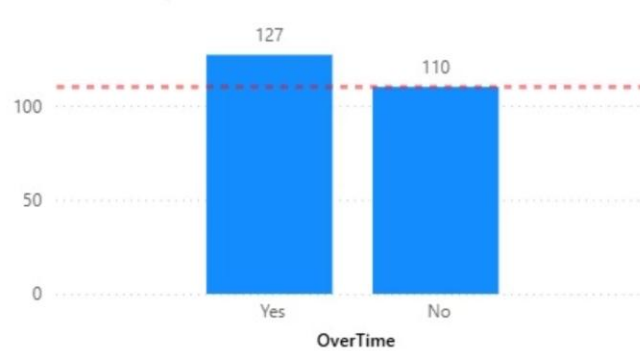
Avg Job Satisfaction by YearsAtCompany



Avg Monthly Income by TotalWorkingYears



Attrition Count by OverTime



7.80K
Projected Income

14.51%
Projected Attrition Reduction

213
Employees At Risk After Raise

8.13K
Projected Income After Experience

otion	Attrition Risk Score	Risk Category
0	100	High Risk
0	100	High Risk
2	100	High Risk
1	90	High Risk
9	90	High Risk
5	90	High Risk
7	90	High Risk
0	85	High Risk
1	85	High Risk
0	85	High Risk
0	85	High Risk
2	85	High Risk
13	85	High Risk
0	85	High Risk
7	85	High Risk
5	85	High Risk
0	85	High Risk
0	85	High Risk
0	85	High Risk
2	85	High Risk
0	85	High Risk
0	80	High Risk
2	80	High Risk
2	80	High Risk
2	80	High Risk
0	80	High Risk
1	80	High Risk
1	80	High Risk
10	80	High Risk
1	80	High Risk
0	75	High Risk
0	75	High Risk
1	75	High Risk
1	75	High Risk
7	75	High Risk
45		Medium Risk

Tableau Dashboard

Interactive Dashboard Development using Tableau

Key Features:



Dynamic Charts & Filters

Interactive visualizations for real-time data exploration



KPI Monitoring

Total Employees (1,470), Total Attrition (237), Attrition Rate (16.12%), Active Employees (1,233), Avg Age (37)



Department Analysis

Department-wise attrition breakdown and trends



Demographic Insights

Employee segmentation by age group, gender, and job satisfaction



User-Friendly Design

Clear, visually appealing format optimized for end-user engagement

Dashboard Capabilities:

- Drill-down functionality for detailed analysis
- Real-time data updates and trend monitoring
- Comparative analysis across departments and demographics
- Data storytelling through intuitive visualizations
- Interactive filters for custom reporting and insights



HR Analytics Dashboard

KPI	Att.Gender	D.Attrition
E.Attrition	Jop S.	N.E.Age
Att.Age group		

Education Field
☒ Human Resources
☒ Life Sciences
☒ Marketing
☒ Medical
☒ Other

Σ Total Employees

1,470

➡ Total Attrition

237

⚙️ Attrition Rate

16.12%

🔌 Activate Employees

1,233

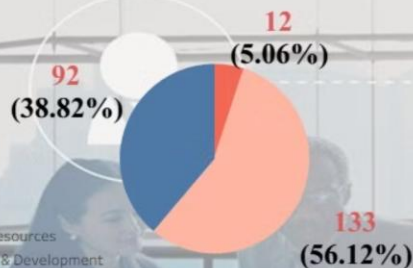
👤 Avg. Age

37

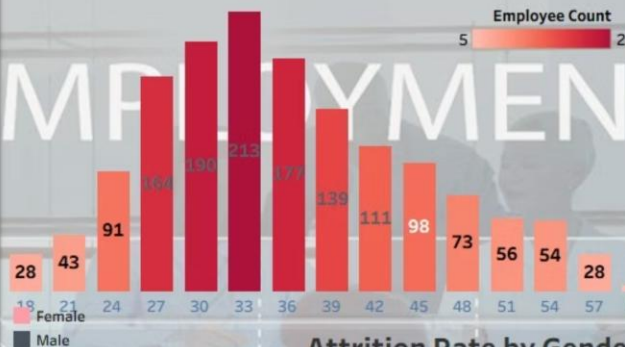
Attrition by Gender



Department Wise Attrition



No of Employee by Age Group



Jop Satisfaction Rating

Job Role	Job Satisfaction				Total
	1	2	3	4	
Human Resources	10	16	13	13	52
Research Director	15	16	27	22	80
Sales Representative	12	21	27	23	83
Manager	21	21	27	33	102
Healthcare Representative	26	19	43	43	131
Manufacturing Director	26	32	49	38	145
Laboratory Technician	56	48	75	80	259
Research Scientist	54	53	90	95	292
Sales Executive	69	54	91	112	326
Total	289	280	442	459	1,470

Education Field Wise Attrition



Attrition Rate by Gender for different Age Group

